

- II. Product of two digits of a number is 16 and quotient of two digits is 4.

15. What is the value of the two-digit number?

(Bank P.O., 2009)

- I. The product of the digits is 72 and the difference between the digits is 1.

- II. The digit at the unit place is greater than the other.

**Directions (Questions 16 to 22): Each of the questions given below consists of a question followed by three statements. You have to study the question and the statements and decide which of the statement(s) is/are necessary to answer the given question.**

16. What is the two-digit number?

(M.B.A., 2002)

- I. Sum of the digits is 7.

- II. Difference between the number and the number obtained by interchanging the digits is 9.

- III. Digit in the ten's place is bigger than the digit in the unit's place by 1.

- (a) I and II only (b) II and III only  
(c) I and III only (d) All I, II and III  
(e) None of these

17. What is the sum of the digits of the two-digit number?

- I. The ratio between the ten's digit and unit's digit of the number is 3 : 2.

- II. The number obtained on revering the order of its digits is 18 less than the original number.

- III. The product of the digits is 24.

- (a) Any two of the three (b) I only or II and III only  
(c) All I, II and III (d) I and II only  
(e) None of these

18. What will be the sum of two numbers?

- I. Among the two numbers, the bigger number is greater than the smaller number by 6.

- II. 40% of the smaller number is equal to 30% of the bigger number.

- III. The ratio between half of the bigger number and one-third of the smaller number is 2 : 1.

- (a) I and II only

- (b) II and III only

- (c) All I, II and III

- (d) Any two of the three

- (e) None of these

19. What is the two-digit number?

(M.A.T., 2005)

- I. The difference between the two-digit number and the number formed by interchanging the digits is 27.

- II. The difference between the two digits is 3.

- III. The digit at unit's place is less than that at ten's place by 3.

- (a) I and II only (b) I and III only  
(c) All I, II and III (d) I, and either II or III  
(e) Even with all I, II and III, answer cannot be given.

20. What is the two-digit number?

(Bank P.O., 2004)

- I. Digit in the ten's place is cube of the digit in unit's place.

- II. Digit in the ten's place is four times the digit in the unit's place.

- III. The two digits are not equal.

- (a) I and II only (b) I and III only  
(c) Any two of the three (d) I and either II or III only  
(e) None of these

21. Find three positive consecutive even numbers.

(M.A.T., 2006)

- I. The average of four consecutive even numbers starting from the last of the given numbers is 17.

- II. The difference of the highest and the lowest numbers is 4.

- III. The sum of the squares of the three numbers is 440.

- (a) I only (b) I and II only  
(c) III only (d) Either I or III

22. What is the two-digit number?

(Bank P.O., 2006)

- I. The number obtained by interchanging the digits of the number is greater than the original number by 18.

- II. Sum of the digits of the number is 14.

- III. Difference between the two digits of the number is 2.

- (a) Any two of the three (b) Only I and III  
(c) II and either I or III (d) III and either I or II  
(e) All of these

## ANSWERS

1. (a) 2. (d) 3. (e) 4. (c) 5. (e) 6. (e) 7. (b) 8. (a) 9. (e) 10. (b)  
11. (a) 12. (a) 13. (e) 14. (b) 15. (e) 16. (e) 17. (a) 18. (e) 19. (e) 20. (a)  
21. (d) 22. (c)

## SOLUTIONS

1. I. Two digits with difference 9 are 9 and 0. And, the two-digit number formed using 9 and 0 is 90.  
II. Let the ten's and unit's digits be  $x$  and  $y$  respectively. Then,  $x - y = x + y \Leftrightarrow 2y = 0 \Leftrightarrow y = 0$ . So, the ten's digit cannot be determined. Thus, I only gives the answer, while II does not.  $\therefore$  The correct answer is (a).

2. I. gives,  $a - b = 2$  ... (i)  
or  $b - a = 2$  ... (ii)  
II. gives,  $a + b = 4$  ... (iii)  
Solving (i) and (iii), we get :  $a = 3$  and  $b = 1$ .  
Solving (ii) and (iii), we get :  $a = 1$  and  $b = 3$ .  
So, the required number is either 13 or 31.  
Thus, even both I and II together do not give the answer.  $\therefore$  The correct answer is (d).